5-12-19 (E)

**K. J. Somaiya College of Engineering, Mumbai-77**
**(Autonomous College Affiliated to University of Mumbai)**

**End Semester Exam**
Nov - Dec 2019

**Max. Marks: 100** | **Duration: 3 hrs**
Class: SY BTECH | Semester: III
Name of the Course: Theory of Computers | Branch: IT
Course Code: UITC 305

**Instructions:**
**(1) All Questions are Compulsory**
**(2) Draw neat diagrams**
**(3) Assume suitable data if necessary**

<table>
<tr><th>Question No.</th><th></th><th>Max. Marks</th></tr>
<tr><td>Q 1 (a)</td><td>Define following-<br>1. Finite State Machine<br>2. Ambiguous grammar<br>3. Left most derivation<br>4. Regular grammar (type 3)</td><td>10</td></tr>
<tr><td>Q 1 (b)</td><td>Explain with example the process of converting Mealy machine to its corresponding Moore machine.</td><td>10</td></tr>
<tr><td>Q2 (a)</td><td>Explain NFA to DFA conversion method and Convert given table to an NFA diagram and then to an equivalent DFA.<br><table><tr><td>State \Input</td><td>0</td><td>1</td></tr><tr><td>$q_0$</td><td>$\{q_0, q_1\}$</td><td>$\{q_1\}$</td></tr><tr><td>$q_1$</td><td>$\{\}$</td><td>$\{q_0, q_1\}$</td></tr></table></td><td>10</td></tr>
<tr><td>Q2 (b)</td><td>Give Regular Expressions and construct finite automata for-<br>1. Binary strings containing at least one 11 & at least one 00<br>2. Strings with even number of a's</td><td>10</td></tr>
<tr><td>Q3 (a)</td><td>Consider the grammar $G = \{(S, A, B), (0, 1), P, S\}$, where P consists of:<br>1. $S \rightarrow A1B$    2. $A \rightarrow 0A \mid \epsilon$    3. $B \rightarrow 0B \mid 1B \mid \epsilon$<br>Show the leftmost and rightmost derivation for the input strings i) 00101 ii) 00011<br>Is given G Ambiguous?.<br><br>OR<br><br>Find whether the string 'aabbb' is a member of L(G), where G is defined as:<br>$S \rightarrow XY$<br>$X \rightarrow YY \mid a$<br>$Y \rightarrow XY \mid b$<br>Is given G Ambiguous?<br>Give leftmost & rightmost derivation for the strings<br>i) aaaabbbb<br>ii) aaaabb</td><td>10</td></tr>
<tr><td>Q3 (b)</td><td>Write an algorithm to convert any CFG to its equivalent CNF. Using same algorithm convert $A \rightarrow abA \mid baA \mid \epsilon$<br><br>OR<br><br>Find context-free grammars generating each of the following languages:<br>1. $L1 = \{ a^i b^j c^k \mid j = i + k\}$<br>2. $L2 = \{ a^i b^j c^k \mid i = j \text{ or } j = k\}$</td><td>10</td></tr>
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| Q4 (a) | Design PDA that checks for well-formed parentheses.<br><br>OR<br><br>Discuss the relative powers of FSM, PDM and Turing machine (TM). | 10 |
| Q4 (b) | Convert the following grammar to Greibach normal form (GNF):<br>$S \rightarrow B\ S$; $S \rightarrow A\ a$; $A \rightarrow b\ c$; $B \rightarrow A\ c$ | 10 |
| Q5 (a) | Design a TM that replaces all occurrences of '111' by '101' from a sequence of 0's & 1's | 10 |
| Q5 (b) | Write short note on Universal TM | 10 |

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|---|---|---|
| Q4(b) | OR<br>Convert the given CFG to equivalent Greibach Normal Form (GNF)<br>$S \rightarrow 0A0 \mid 1B1 \mid BB$<br>$A \rightarrow C$<br>$B \rightarrow S \mid A$<br>$C \rightarrow S \mid \epsilon$ | 10 |